

SP-4 Phase D — Multi-Provider Parallel Search SDK (implementation plan)

Plan date: 2026-05-13 **Design:** docs/superpowers/specs/2026-05-13-sp4-phase-d-design.md **Parent SP:** docs/superpowers/specs/2026-05-12-sp4-multi-provider-redesign.md **Prerequisite:** Phase C complete (HEAD 6db79c50).

6 tasks, ~20 steps. The SDK-side work (Tasks 1-3) + ActiveTrackerSection removal (Task 4) are autonomously executable. The Search-Result UI branch (Task 5) is also autonomous. Cancellation + backpressure unit tests (Task 6) ship in the same commit. Compose UI Challenges C32 + C33 are documented as KDoc rehearsal protocols; operator runs them on the gating emulator before next tag.

Task 1 — MultiSearchEvent sealed hierarchy + builder

The event vocabulary the streaming SDK method emits.

Files:

- Create: core/tracker/client/src/main/kotlin/lava/tracker/client/MultiSearchEvent.kt

Steps:



Step 1: Sealed hierarchy.

```
sealed interface MultiSearchEvent {
    val providerId: String

    data class ProviderStart(
        override val providerId: String,
        val displayName: String,
    ) : MultiSearchEvent

    data class ProviderResults(
        override val providerId: String,
        val items: List<TorrentItem>,
    ) : MultiSearchEvent

    data class ProviderFailure(
```

```

        override val providerId: String,
        val reason: String,
        val cause: Throwable? = null,
    ) : MultiSearchEvent

    data class ProviderUnsupported(
        override val providerId: String,
    ) : MultiSearchEvent

    data class AllProvidersDone(
        override val providerId: String = "",
        val unified: UnifiedSearchResult,
    ) : MultiSearchEvent
}

```



Step 2: Builder helper. Add a companion fun

`UnifiedSearchResult.Companion.fromEvents(query, page, events): UnifiedSearchResult` — consume the event sequence, build the provider statuses + resultsByProvider maps, run `DeduplicationEngine.deduplicate(...)`, return the assembled `UnifiedSearchResult`.



Step 3: Compile.

Task 2 — Rewrite `LavaTrackerSdk.multiSearch` to parallel fan-out

Files:

- Modify: `core/tracker/client/src/main/kotlin/lava/tracker/client/LavaTrackerSdk.kt`

Steps:



Step 1: Extract per-provider logic into a private suspend fun `runOneProvider(id, descriptor, request, page): ProviderOutcome` returning a sealed result (`Success(items)`, `Failure(reason)`, `Unsupported`, `NotRegistered`).



Step 2: Rewrite `multiSearch` body:

```

suspend fun multiSearch(
    request: SearchRequest,
    providerIds: List<String>,
    page: Int = 0,
): UnifiedSearchResult = coroutineScope {
    val outcomes = providerIds.map { id ->
        async { id to runOneProvider(id, request, page) }
    }
}

```

```

    }.awaitAll()
    // Build UnifiedSearchResult from the parallel outcomes (logic
    // moved here from current sequential impl)
    val statuses = outcomes.map { (id, outcome) ->
        outcome.toStatus(id) }
    val resultsByProvider = outcomes.mapNotNull { (id, outcome) ->
        (outcome as? ProviderOutcome.Success)?.let { id to
            it.items }
    }.toMap()
    val displayNames = outcomes.associate { (id, outcome)
        -> id to outcome.displayName(id) }
    val deduped =
        DeduplicationEngine.deduplicate(resultsByProvider,
            displayNames)
    UnifiedSearchResult(
        query = request.query,
        page = page,
        items = deduped,
        totalPages = 1,
        providerStatuses = statuses,
    )
}

```



Step 3: Compile.

Task 3 — Add streamMultiSearch(...) via channelFlow

Files:

- Modify: core/tracker/client/src/main/kotlin/lava/tracker/client/LavaTrackerSdk.kt

Steps:



Step 1: Signature + body.

```

fun streamMultiSearch(
    request: SearchRequest,
    providerIds: List<String>,
    page: Int = 0,
): Flow<MultiSearchEvent> = channelFlow {
    val statuses = mutableMapOf<String, ProviderSearchStatus>()
    val resultsByProvider = mutableMapOf<String,
        List<TorrentItem>>()
    val displayNames = mutableMapOf<String, String>()

    for (id in providerIds) {
        launch {

```

```

        val descriptor = registry.list().firstOrNull {
            it.trackerId == id }
        if (descriptor == null) {
            send(MultiSearchEvent.ProviderFailure(id, "not
registered"))
            statuses[id] = ProviderSearchStatus(id, id,
ProviderSearchState.FAILURE, errorMessage = "not
registered")
            return@launch
        }
        displayNames[id] = descriptor.displayName
        send(MultiSearchEvent.ProviderStart(id,
descriptor.displayName))

        if (TrackerCapability.SEARCH !in
descriptor.capabilities) {
            send(MultiSearchEvent.ProviderUnsupported(id))
            statuses[id] = ProviderSearchStatus(id,
descriptor.displayName, ProviderSearchState.UNSUPPORTED)
            return@launch
        }

        val client = registry.get(id, MapPluginConfig())
        val feature =
client.getFeature(SearchableTracker::class)
        if (feature == null) {
            send(MultiSearchEvent.ProviderUnsupported(id))
            statuses[id] = ProviderSearchStatus(id,
descriptor.displayName, ProviderSearchState.UNSUPPORTED)
            return@launch
        }
        try {
            val result = feature.search(request, page)
            resultsByProvider[id] = result.items
            statuses[id] = ProviderSearchStatus(id,
descriptor.displayName, ProviderSearchState.SUCCESS,
resultCount = result.items.size)
            send(MultiSearchEvent.ProviderResults(id,
result.items))
        } catch (t: Throwable) {
            statuses[id] = ProviderSearchStatus(id,
descriptor.displayName, ProviderSearchState.FAILURE,
errorMessage = t.message ?: "search failed")
            send(MultiSearchEvent.ProviderFailure(id,
t.message ?: "search failed", t))
        }
    }
}

// Wait for all launches to complete; channelFlow's
// `awaitClose` is not needed
// because the `launch`-es are children of this scope and
// complete before
// channelFlow returns.
}.onCompletion { cause ->

```

```

    if (cause == null) {
        val deduped =
            DeduplicationEngine.deduplicate(resultsByProvider,
                displayNames)
        val unified = UnifiedSearchResult(
            query = request.query,
            page = page,
            items = deduped,
            totalPages = 1,
            providerStatuses = statuses.values.toList(),
        )
        emit(MultiSearchEvent.AllProvidersDone(unified = unified))
    }
}

```

Note: the local mutable maps captured by the inner launches are safe in coroutine-scope mutation (single-threaded confined to the parent's dispatcher) because `channelFlow` runs the producer on a single dispatcher unless the user opts into `flowOn`. If we ever move to `flowOn(Dispatchers.IO.limitedParallelism(...))`, swap the maps for `ConcurrentHashMap`.



Step 2: Compile.

Task 4 — Delete `ActiveTrackerSection` + related state/action

The Phase C-added affordance is dead post-Phase-D. Delete with `@Deprecated` markers on the SDK API surface it called.

Files:

- Delete: `feature/provider_config/src/main/kotlin/lava/provider/config/sections/ActiveTrackerSection.kt`
- Modify: `feature/provider_config/src/main/kotlin/lava/provider/config/ProviderConfigState.kt` — remove `activeTrackerId` field
- Modify: `feature/provider_config/src/main/kotlin/lava/provider/config/ProviderConfigAction.kt` — remove `MakeActive`
- Modify: `feature/provider_config/src/main/kotlin/lava/provider/config/ProviderConfigViewModel.kt` — remove the `MakeActive` branch + the `activeTrackerId` field assignment in `onCreate`
- Modify: `feature/provider_config/src/main/kotlin/lava/provider/config/ProviderConfigScreen.kt` — remove `ActiveTrackerSection` import + invocation
- Modify: `core/tracker/client/src/main/kotlin/lava/tracker/client/LavaTrackerSdk.kt` — mark `activeTrackerId()` and both `switchTracker` overloads `@Deprecated("Use multiSearch/streamMultiSearch – single-active-tracker semantics removed by SP-4 Phase D")`.

Steps:



Step 1: `git rm ActiveTrackerSection.kt`.



Step 2: Edit each of the 5 modified files to drop the corresponding lines.



Step 3: Compile both feature modules.

Task 5 — `SearchResultViewModel` consumes `streamMultiSearch` when API is not configured

Files:

- Modify: `feature/search_result/src/main/kotlin/lava/search/result/SearchResultViewModel.kt`

Steps:



Step 1: Branch in `onCreate`.

```
onCreate = {  
    observeFilter()  
    when {  
        mutableFilter.value.providerIds == null ->  
            observePagingData()  
        currentEndpointIsGoApi() ->  
            observeSseSearch(mutableFilter.value)  
        else -> observeStreamMultiSearch(mutableFilter.value)  
    }  
}
```



Step 2: Helper `currentEndpointIsGoApi()` reads `observeSettingsUseCase().first().endpoint` is `Endpoint.GoApi`.



Step 3: New private `observeStreamMultiSearch(filter)` consumes `sdk.streamMultiSearch(...)`. On each event:

- `ProviderStart(id, name)` → reduce `searchContent.activeProviders` adding/updating the provider row to `SEARCHING`.
- `ProviderResults(id, items)` → append to `searchContent.items` + mark provider `DONE`.
- `ProviderFailure(id, reason, _)` → mark provider `ERROR(reason)`.
- `ProviderUnsupported(id)` → mark provider `UNSUPPORTED`.
- `AllProvidersDone(unified)` → final reduce with the deduplicated unified result.

Reuse the existing `handleSseEvent`-style state-reduce logic; refactor the shared parts into a private helper `applyProviderEvent(state, event): SearchPageState`.



Step 4: Compile.

Task 6 — Cancellation + backpressure unit tests

Files:

- Create: `core/tracker/client/src/test/kotlin/lava/tracker/client/LavaTrackerSdkParallelSearchTest.kt`

Steps:



Step 1: Parallel fan-out timing test. Fake 3 providers with 500ms latency each. Assert `multiSearch(...)` completes in $< 700\text{ms}$ (max latency + overhead), not 1500ms (sum of latencies).



Step 2: Cancellation test. Start `streamMultiSearch(...)` collection; cancel after first `ProviderStart` event arrives; assert remaining provider coroutines are cancelled (verify by injecting a fake `SearchableTracker` whose search is a `suspendCancellableCoroutine` and asserting the cancellation handler ran).



Step 3: Backpressure test. Inject 5 fake providers that each emit instantly; use a collector with `delay(50)` between collects; assert all `ProviderStart` + `ProviderResults` + `AllProvidersDone` events are received in order (no drops, no reorderings).



Step 4: Bluff-Audit rehearsal.

- For (1): Mutate `multiSearch` body to `for (id in providerIds) { runOneProvider(...) }` (sequential). Confirm test fails with "took 1500ms, expected $< 700\text{ms}$ ".
- For (2): Remove the `coroutineScope { }` wrapper so children are unstructured. Confirm cancellation test fails — child coroutines outlive the parent cancellation.
- For (3): Change `channelFlow` buffer to `Channel.UNLIMITED` (no backpressure). Confirm test still passes for this scenario but document that the change weakens the contract; the test's purpose is to lock the `SUSPEND` policy. (For a falsifiable test of backpressure ITSELF, mutate the collector to `NOT delay` and assert that the emissions happen before the delay — but this is testing the test, not the SUT. Stick with the order-preservation assertion as the primary signal.)

Record all three mutations + observed-failures in commit body.



Step 5: Compile + run tests. ./

gradlew :core:tracker:client:testDebugUnitTest.

Task 7 — CONTINUATION update + commit + push

Steps:



Step 1: Update CONTINUATION.md §0 with Phase D delivery.



Step 2: Local CI gate. bash scripts/check-constitution.sh + the spotless/test runs.



Step 3: Commit with:

- Bluff-Audit stamps for LavaTrackerSdkParallelSearchTest.kt (3 mutations).
- Notes on @Deprecated-marked API surfaces and the migration target.
- Co-Authored-By trailer.



Step 4: Push to github + gitlab. Verify SHA convergence.

Risk register

Risk	Mitigation
channelFlow's send from inner launch can deadlock if the collector is slow + the producer holds a mutex	Avoid mutex inside the producer; rely on coroutine-confinement of the local maps (single dispatcher). Tested by Task 6 Step 3.
The legacy paging path still calls LavaTrackerSdk.activeTrackerId()	Deprecation marker is non-breaking; legacy callers continue to compile. Full removal is a future phase post-paging-migration.
streamMultiSearch emits AllProvidersDone via onCompletion — if the collector cancels mid-stream, AllProvidersDone is NOT emitted	Intentional. The consumer's observeStreamMultiSearch ViewModel logic does not depend on AllProvidersDone for final state; it incrementally reduces on each ProviderResults event. AllProvidersDone is the "all OK, here's the deduplicated summary" event for non-streaming callers.
Phase D commit and Phase C commit both reference ActiveTrackerSection — risk of merge surprise	Phase D ships only on master, ahead of Phase C in linear history. No branching surprise.
Cancellation test may flake on slow CI	

Risk	Mitigation
	Use generous timeouts (1000ms cancellation-propagation window).

Deliverable summary

A single commit (or 2-3 if grouping by domain) that:

1. Adds `MultiSearchEvent` + `parallel multiSearch` + `streamMultiSearch` to `LavaTrackerSdk`.
2. Wires `SearchResultViewModel` to consume the new flow when the Go API is not configured.
3. Deletes `ActiveTrackerSection` + related state/action; marks `LavaTrackerSdk.activeTrackerId()` + `switchTracker()` `@Deprecated`.
4. Adds 3 unit tests (parallel-timing, cancellation, backpressure) with Bluff-Audit stamps.
5. Documents C32 + C33 as Compose UI Challenges with KDoc rehearsal protocols (operator-bound execution).
6. Updates `CONTINUATION.md`.
7. Lands on both mirrors with verified SHA convergence.

Estimated session time: 1-2 hours of execution + the three unit-test mutation rehearsals (~3 minutes each). The Compose UI Challenges C32 + C33 are owed for operator execution before next tag, not for this implementation commit.